

1.a

(1)

Show that $\arg \min_{u: u^T u = 1} \sum_{i=1}^n \|x^{(i)} - \mu(x^{(i)})\|_2^2$

is equivalent to the optimization problem.

$$\arg \max_{u: u^T u = 1} u^T \left(\sum_{i=1}^n x^{(i)} (x^{(i)})^T \right) u.$$

Given u is unit length vector, $\{x^{(1)}, \dots, x^{(n)}\}$ has zero mean and unit-variance $N(0, 1)$, $V = \alpha u$; $\alpha \in \mathbb{R}$; $f_u(x) = \arg \min_{V \in \mathbb{R}} \|x - V\|_2^2$

Solution:-

$$f_u(x) = \arg \min_{V \in \mathbb{R}} \|x - V\|_2^2; V = \alpha u; u^T u = 1$$

where u is unit vector.

Substituting $V = \alpha u$ in $f_u(x)$

$$f_u(x) = \arg \min_{\alpha \in \mathbb{R}} \|x - \alpha u\|_2^2$$

$$\begin{aligned} \|x - \alpha u\|_2^2 &= (x - \alpha u)^T (x - \alpha u) \\ &= x^T x - x^T \alpha u - u^T \alpha^T x + \alpha^T u^T u \\ &= x^T x - 2\alpha u^T x + \alpha^T u^T u \\ &= x^T x - 2\alpha u^T x + \alpha^T \quad [u^T u = 1] \end{aligned}$$

As this is in Quadratic form over α , to minimize, let's set the derivative to zero

$$\frac{d}{d\alpha} (\alpha^T - 2\alpha u^T x + x^T x) = 0.$$

$$2\alpha - 2u^T x = 0$$

$$u^T x = \alpha \Rightarrow \boxed{\alpha = u^T x} \text{ minimizes } f_u(x).$$

1.0

$$\therefore f_v(x) = (v^T x) \cdot v$$

(2)

Now consider $\arg \min_{v: v^T v = 1} \sum_{i=1}^n \|x^{(i)} - f_v(x^{(i)})\|_2^2$

$$\arg \min_{v: v^T v = 1} \sum_{i=1}^n \left[\|x^{(i)}\|^2 - 2 (v^T x^{(i)})^2 + (v^T x^{(i)})^2 \|v\|^2 \right]$$

$$= \arg \min_{v: v^T v = 1} \sum_{i=1}^n \left[\|x^{(i)}\|^2 - (v^T x^{(i)})^2 \right]$$

$$= \arg \min_{v: v^T v = 1} \left[\sum_{i=1}^n \|x^{(i)}\|^2 - \sum_{i=1}^n (v^T x^{(i)})^2 \right]$$

In order to minimize $\sum_{i=1}^n \|x^{(i)} - f_v(x^{(i)})\|_2^2$, we need to maximize the term after the minus (negative) symbol. Note $\|x^{(i)}\|^2$ is independent of v .

$$\therefore \arg \min_{v: v^T v = 1} \sum_{i=1}^n \|x^{(i)} - f_v(x^{(i)})\|_2^2$$

$$\cong \arg \max_{v: v^T v = 1} \sum_{i=1}^n (v^T x^{(i)})^2$$

$$= \arg \max_{v: v^T v = 1} \sum_{i=1}^n v^T (x^{(i)} x^{(i)T}) v$$

$$\arg \min_{v: v^T v = 1} \sum_{i=1}^n \|x^{(i)} - f_v(x^{(i)})\|_2^2 = \arg \max_{v: v^T v = 1} \sum_{i=1}^n v^T (x^{(i)} x^{(i)T}) \cdot v$$

has been proved, i.e., the unit length vector v that minimizes the mean squared error between projected points and original points corresponds to the first principal component.

1.5.i $M^T M$ is real, Symmetric & Square matrix. write its eigen decomposition in terms of $Q \Lambda Q^T$

Solution

If M is real; M^T is also real, therefore

$M^T M$ is also real as it involve multiplying and summing the real numbers within M and M^T .

$\therefore \underline{M^T M \text{ is real.}}$

For a matrix A to be Symmetric; $A^T = A$.

for $M^T M$ to be Symmetric; $(M^T M)^T = M^T M$. (should be)

$$(M^T \cdot M)^T = M^T \cdot (M^T)^T = M^T \cdot M.$$

$\therefore M^T \cdot M = (M^T \cdot M)^T$, i.e. $M^T M$ is Symmetric.

If M is a ~~square~~ matrix of size $p \times q$, M^T is $q \times p$.
therefore $M^T M$ is of size $q \times q$, which is a
Square matrix.

As $M^T M$ is a Symmetric, real, Square matrix; it
can be written in Eigen decomposition as

$$M^T M = Q \Lambda Q^T \text{ where } Q^T \cdot Q = I$$

Q is the matrix of Eigen vectors. and $\Lambda = \text{diag}(\lambda_1, \dots, \lambda_q)$

~~then~~ ~~So~~ ~~representation~~ ~~matrix~~ ~~argument~~ ~~matrix~~ ~~matrix~~

$$M^T M = Q \Lambda Q^T$$

1.b.ii

$$M = U \Sigma V^T; \quad U \in \mathbb{R}^{p \times p}, V \in \mathbb{R}^{q \times q} \quad \Sigma \in \mathbb{R}^{p \times q} \quad (9)$$

$$M^T M = (U \Sigma V^T)^T (U \Sigma V^T)$$

$$M^T M = V \Sigma^T \underbrace{U^T U}_{=I} \Sigma V^T \quad [\text{we know } U^T U = I]$$

$$M^T M = V \Sigma^T \Sigma V^T$$

$$\boxed{M^T M = V \Sigma^2 V^T}$$

where V is Matrix of Eigen vectors of $M^T M$. and
 Σ^2 is a diagonal matrix i.e. $\text{diag}(\sigma_1^2, \dots, \sigma_q^2)$
and Eigen values $(\lambda_1, \dots, \lambda_q)$ are $(\sigma_1^2, \dots, \sigma_q^2)$ respectively

1.b.iii Relationship Between the Eigen values of $M^T M$ and the Singular values of M .

Comparing 1.b.ii and 1.b.i above

$V \Sigma^2 V^T$ is in the form $Q \Lambda Q^T$.

where $Q = V$; $\Lambda = \Sigma^2$ where $\Sigma = \begin{bmatrix} \sigma_1 & & \\ & \sigma_2 & \\ & & \ddots \\ & & & \sigma_q \end{bmatrix}$

We can say, the Eigen values of the $M^T M$ are simply the Squares of the Singular values of the data matrix M .

$$\text{i.e. } \lambda_i = \sigma_i^2 \quad \forall i = 1 \text{ to } q.$$

where λ_i represent Eigen values of $M^T M$,
 σ_i represent Singular values of M .

2.2

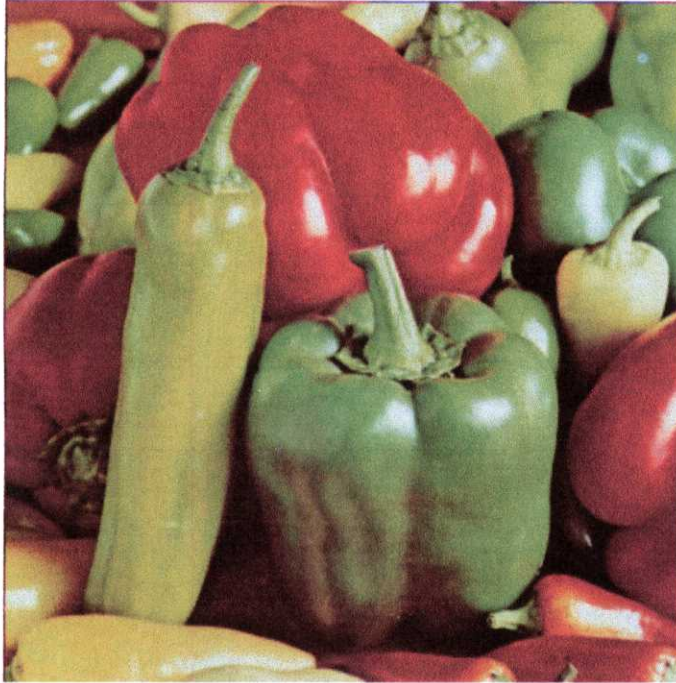
5

Output of ls -la command to show the respective sizes in bytes

-rw-r--r--@ 1 dhruvakhagi staff 293404 Mar 1 23:12 orig_large.png

-rw-r--r--@ 1 dhruvakhagi staff 187792 Mar 1 23:12 updated_large.png

Original large image



Updated large image



Q.6

(6)

Compression factor

As per the output of ls -ltr Linux Command;

Size of original large image (orig-large.png) - 293404 Bytes

Size of updated large image (updated-large.png) - 187792 Bytes.

$$\text{Compression factor} = \frac{\text{original size}}{\text{compressed size}} = \frac{293404}{187792}$$

$$\boxed{\text{Compression factor} = 1.562388}$$

i.e. the original large image is 1.56 larger than the updated compressed large image.

3.00 ①
prove $D_{KL}(P||Q) \geq 0$ and $D_{KL}(P||Q) = 0$ if & only if $P = Q$.

Solution:-

Expected value of a function $f(x)$ for a discrete random variable X with probability distribution $P(x)$ is defined as

$$E_P[f(x)] = \sum_x f(x) \cdot P(x).$$

$$\text{Let } \frac{Q(x)}{P(x)} = f(x).$$

$$\therefore E_P[f(x)] = \sum_x P(x) \cdot \frac{Q(x)}{P(x)} = \sum_x Q(x) = 1$$

$$\therefore \log\left(\sum_x P\left(\frac{Q(x)}{P(x)}\right)\right) = \log 1 = 0 \quad \text{--- (1)} \quad \left(\begin{array}{l} \text{as summation} \\ \text{of probabilities } P(x) \\ \text{equals 1} \end{array}\right)$$

Jensen's inequality states, for a concave function $g(x)$,

$$E[g(x)] \leq g[E(x)].$$

Since $\log$ is concave function, applying Jensen's inequality

we get
$$\log\left(\sum_x P\left(\frac{Q(x)}{P(x)}\right)\right) \geq \sum_x P\left(\log\left(\frac{Q(x)}{P(x)}\right)\right)$$

from (1) we know
$$\log\left(\sum_x P\left(\frac{Q(x)}{P(x)}\right)\right) = 0.$$

$$\therefore E_P\left(\sum_x \log\left(\frac{Q(x)}{P(x)}\right)\right) \leq 0$$

$$E_P \left(\sum_n \log \left(\frac{Q(x)}{P(x)} \right) \right) \leq 0.$$

By the definition of Expectation function E_P .

$$\sum_n P(x) \cdot \log \left(\frac{Q(x)}{P(x)} \right) \leq 0.$$

$$\sum_n P(x) \cdot \log \left(\frac{P(x)}{Q(x)} \right) \geq 0$$

By definition $\sum_n P(x) \cdot \log \left(\frac{P(x)}{Q(x)} \right)$

equals to $D_{KL}(P||Q)$.

reason:-
swapping
numerator &
denominator inside
log function

Therefore.

$$D_{KL}(P||Q) = \sum_n P(x) \cdot \log \left(\frac{P(x)}{Q(x)} \right) \geq 0$$

$$\text{ie } D_{KL}(P||Q) \geq 0.$$

has been proved.

3.2

Now proving $D_{KL}(P||Q) = 0$ $\iff$ & only if P & Q are equal
ie $P = Q$. Q

$$D_{KL}(P||Q) = \sum_n P(n) \cdot \log\left(\frac{P(n)}{Q(n)}\right).$$

we know $\sum_n P(n)$ is sum of probabilities
of all n which equals to 1.

therefore in order for $D_{KL}(P||Q)$
to equal to zero, $\log\left(\frac{P(n)}{Q(n)}\right)$ should
equals to zero.

$$\text{ie } \log\left(\frac{P(n)}{Q(n)}\right) = 0.$$

$$\text{ie } \log\left(\frac{P(n)}{Q(n)}\right) = \log(1)$$

$$\text{ie } \frac{P(n)}{Q(n)} = 1$$

$$\text{ie } P(n) = Q(n).$$

$$\text{ie } \boxed{P = Q}$$

Hence $D_K(P||Q) = 0$ if & only if $P = Q$
has been proved.

3.b

(10)

From joint probability distribution we know

$$P(x, y) = P(y/x) \cdot P(x)$$

$$Q(x, y) = Q(y/x) \cdot Q(x)$$

$$D_{KL}(P(x, y) \parallel Q(x, y)) = \sum_{x, y} P(x, y) \cdot \log \left(\frac{P(x, y)}{Q(x, y)} \right)$$

$$= \sum_{x, y} P(x, y) \cdot \log \left[\frac{P(y/x) \cdot P(x)}{Q(y/x) \cdot Q(x)} \right]$$

$$= \sum_{x, y} P(x, y) \left[\log \left(\frac{P(x)}{Q(x)} \right) + \log \left(\frac{P(y/x)}{Q(y/x)} \right) \right]$$

$$= \sum_{x, y} \left(P(x, y) \cdot \log \left(\frac{P(x)}{Q(x)} \right) \right) + \sum_{x, y} P(x, y) \cdot \log \left(\frac{P(y/x)}{Q(y/x)} \right)$$

$$= \sum_{x, y} P(x, y) \cdot \log \left[\frac{P(x)}{Q(x)} \right] + \sum_{x, y} P(x, y) \cdot \log \left[\frac{P(y/x)}{Q(y/x)} \right]$$

$$= \sum_x \left(\sum_y P(x, y) \cdot \log \left(\frac{P(x)}{Q(x)} \right) \right) + \sum_{x, y} P(x) \cdot P(y/x) \cdot \log \frac{P(y/x)}{Q(y/x)}$$

↓

call this term ①

↓

call this term ②

^{3/2} Taking term ①

⑦

$$\sum_x \left(\sum_y p(x, y) \right) \cdot \log \left(\frac{p(x)}{q(x)} \right)$$

$$= \sum_x p(x) \cdot \log \left(\frac{p(x)}{q(x)} \right)$$

Reason:-
Summing over all y's
 $p(x, y) = p(x)$

$$= D_{KL}(P(x) || Q(x))$$

Taking term ②

$$\sum_{x, y} p(x) \cdot p(y/x) \cdot \log \left(\frac{p(y/x)}{q(y/x)} \right)$$

$$= \sum_x p(x) \cdot \sum_y p(y/x) \cdot \log \left(\frac{p(y/x)}{q(y/x)} \right)$$

$$= \sum_x p(x) \cdot D_{KL}(P(y/x) || Q(y/x))$$

$$= 1 \cdot D_{KL}(P(y/x) || Q(y/x))$$

$$= D_{KL}(P(y/x) || Q(y/x))$$

3.2 Plugging the derivations of ① & ② in original equation ⁽¹²⁾

$$D_{KL}(P(X,Y) \parallel Q(X,Y))$$

$$= \sum_x \left(\sum_y P(x,y) \cdot \log \left(\frac{P(x)}{Q(x)} \right) \right)$$

$$+ \sum_{x,y} P(x) \cdot P(y/x) \cdot \log \left[\frac{P(y/x)}{Q(y/x)} \right]$$

$$= D_{KL}(P(X) \parallel Q(X)) + D_{KL}(P(Y/X) \parallel Q(Y/X))$$

ie

$$D_{KL}(P(X,Y) \parallel Q(X,Y)) = D_{KL}(P(X) \parallel Q(X)) + D_{KL}(P(Y/X) \parallel Q(Y/X))$$

has been proved.

3.5 prove $\arg \min_{\theta} D_{KL}(\hat{P} \| P_{\theta}) = \arg \max_{\theta} \sum_{i=1}^n \log P_{\theta}(x^{(i)})$ (12)

Solution

$$D_{KL}(\hat{P} \| P_{\theta}) = \sum_n \hat{P}(x) \cdot \log\left(\frac{\hat{P}(x)}{P_{\theta}(x)}\right)$$

$$\text{Given } \hat{P}(x) = \frac{1}{n} \sum_{i=1}^n \mathbb{1}\{x^{(i)} = x\}$$

$$D_{KL}(\hat{P} \| P_{\theta}) = \sum_n \left[\hat{P}(x) \cdot \log(\hat{P}(x)) - \hat{P}(x) \cdot \log(P_{\theta}(x)) \right]$$

$$= \sum_n \hat{P}(x) \cdot \log(\hat{P}(x)) - \sum_n \hat{P}(x) \cdot \log(P_{\theta}(x))$$

In the above equality, the first term on the right side

$\sum_n \hat{P}(x) \cdot \log(\hat{P}(x))$ is not dependent on θ .

So In the context of what we are embarked to prove this term is not important or irrelevant, rather a constant.

$$\begin{aligned} \therefore D_{KL}(\hat{P} \| P_{\theta}) &= \text{Constant} - \sum_n \hat{P}(x) \cdot \log(P_{\theta}(x)) \\ &= \text{Constant} - \sum_n \left(\frac{1}{n} \sum_{i=1}^n \mathbb{1}\{x^{(i)} = x\} \cdot \log P_{\theta}(x) \right) \end{aligned}$$

$$\textcircled{10}$$

$$\textcircled{3.4} \quad D_{KL}(\hat{P} \parallel P_0) = \text{Constant} - \sum_x \left(\frac{1}{n} \sum_{i=1}^n \mathbb{1}\{x^{(i)} = x\} \log P_0(x) \right)$$

$$= \text{Constant} - \frac{1}{n} \sum_{i=1}^n \sum_x \mathbb{1}\{x^{(i)} = x\} \log P_0(x).$$

Let's focus on $\sum_x \mathbb{1}\{x^{(i)} = x\}$

this means, only when $x^{(i)} = x$; the indicator is 1, otherwise it is zero, for a given i .

so for a given i , the term becomes 1.

$$\therefore D_{KL}(\hat{P} \parallel P_0) = \text{Constant} - \frac{1}{n} \sum_{i=1}^n \log P_0(x^{(i)}).$$

As $\frac{1}{n} \sum_{i=1}^n \log P_0(x^{(i)})$ has a minus (negative) sign in front of it, minimizing $D_{KL}(\hat{P} \parallel P_0)$ is same as maximizing $\frac{1}{n} \sum_{i=1}^n \log(P_0(x^{(i)}))$

therefore

$$\arg \min_Q D_{KL}(\hat{P} \parallel P_Q) = \arg \max_Q \sum_{i=1}^n \log P_Q(x^{(i)})$$

has been proved.

np.random.seed(0)

```
iMac:cartpole dhruvakbhagi$ python3 cartpole.py
```

```
[INFO] Failure number 1
```

```
[INFO] Failure number 2
```

```
[INFO] Failure number 3
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```
.....
```

```
[INFO] Failure number 231
```

```
[INFO] Failure number 232
```

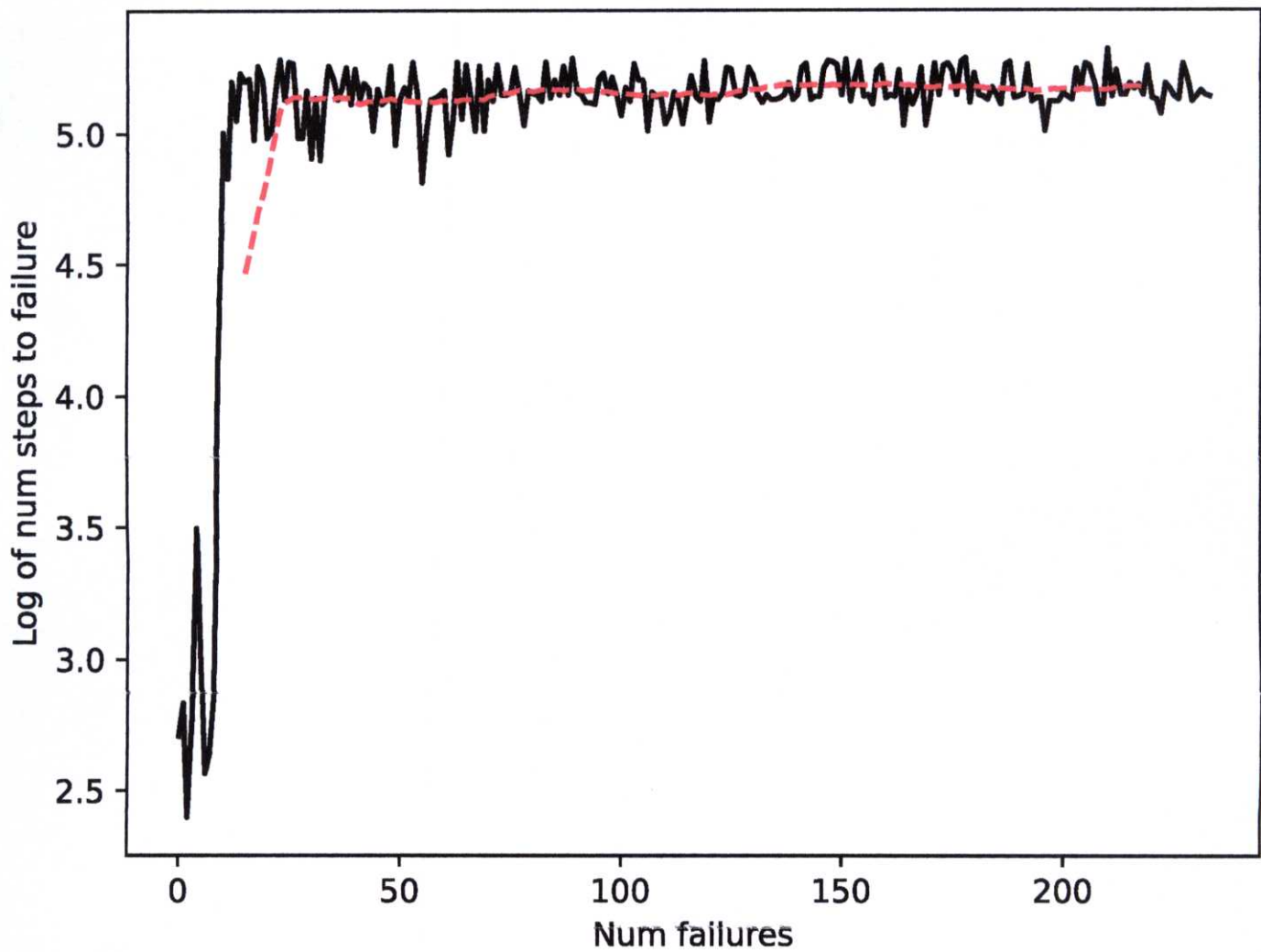
```
[INFO] Failure number 233
```

```
[INFO] Failure number 234
```

- How many trials (# of times pole fall over or cart fall off) did it take before the algorithm converged?

Answer: As per the execution output (copied above), the algorithm took 234 trials before reaching the convergence. ie either pole fall over or cart falling off occurred 234 times before reaching convergence.

np.random.seed(0)



4

17

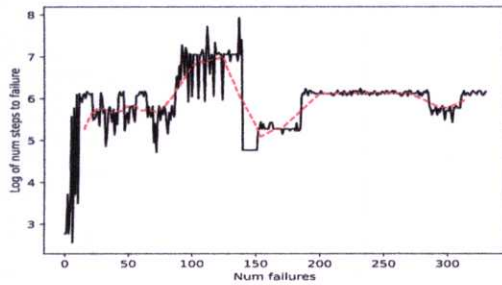
OUTPUT for np.random.seed(1)

iMac:cartpole dhruvakbhagi\$ python3 cartpole.py

[INFO] Failure number 1

.....

[INFO] Failure number 331



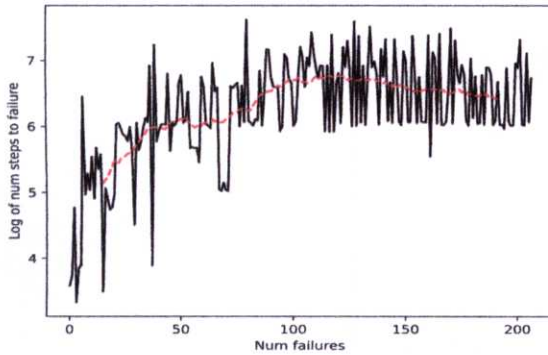
OUTPUT for np.random.seed(2)

iMac:cartpole dhruvakbhagi\$ python3 cartpole.py

[INFO] Failure number 1

.....

[INFO] Failure number 207



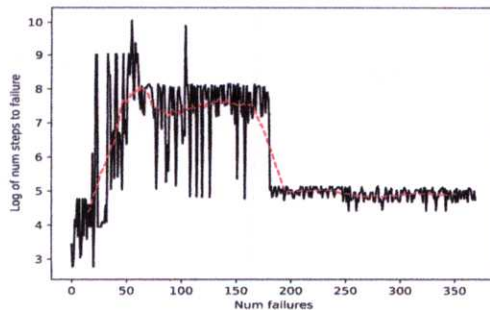
OUTPUT for np.random.seed(3)

iMac:cartpole dhruvakbhagi\$ python3 cartpole.py

[INFO] Failure number 1

....

[INFO] Failure number 370



1. Different seeds required different no. of trials prior to reaching the convergence. The # of trials required is purely random as it boils down to how much luck our chosen random value yields in the initial stages towards maneuvering to the informative state-action pairs.

For example, a luckier random seed value (say zero in our case) required lesser # of trials (234) as initialization resulted in choosing more informative state-action pairs relatively compared to a bit not-so-lucky seed of 3 (in our run) which required lot of exploration in the initial stages thereby needing 370 trials prior to reaching the convergence.

2. Even though different seed values influence the initial learning rate and initial progress, eventually the algorithm learns and reaches the convergence irrespective of the chosen seed value. In other words, a lucky (or) luckier seed value is certainly helpful towards a faster convergence, it's not a hard requirement (or) a blocker for the algorithm.

This indicates the algorithm and its greedy approach is yielding the required results and therefore it is working as expected, irrespective of the chosen seed value. Hence we can say, the algorithm and its approach is robust.